

TOON SHADER DESKTOP



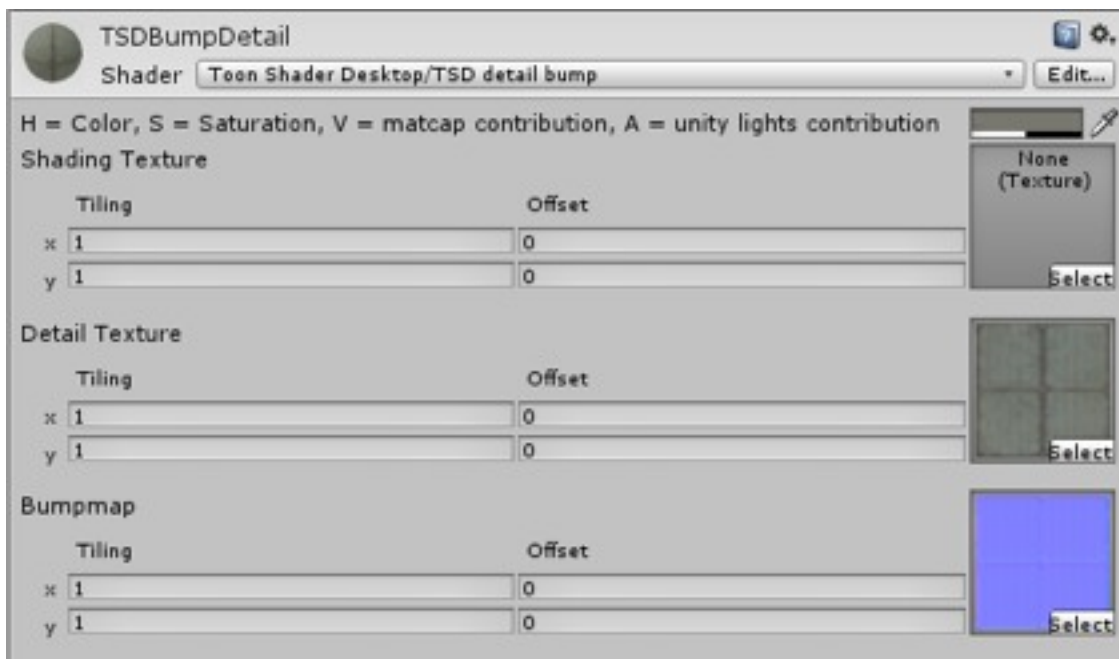
IPPOKRATIS BOURNELLIS

Hi,
Thanks for your interest in Toon Shader Desktop (TSD) !

TSD offers advanced NPR (non photorealistic rendering) in Unity.
Supports light maps, light probes, multiple lights, shadows. Can take advantage of bump maps, if present, for extra shading information.
Outlined and non-outlined variants of all shaders are included. The outlines work for perspective and orthographic cameras and display correct in the Scene view. The thickness and the color of the outlines can be adjusted. I hope TSD will prove useful for your projects !

To better understand the basic principles of using TSD, open the “Toon Bump Detail Outline” scene located in the folder called “scenes”.

Select the “Toon Cube” in the Hierarchy panel and open the TSDBumpDetail in the Inspector panel.

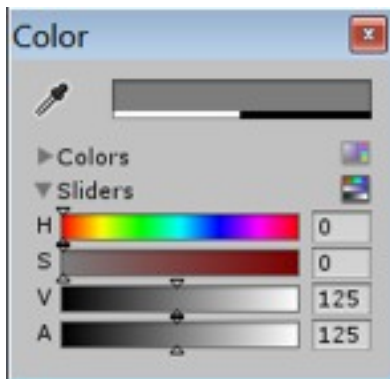


You should now see the material called TSDBumpDetail in the Inspector.
If only the top part of the image shows, click on the sphere on the left of the “Shader” label and the rest of the content will show up.

The “Texture” and the “Bumpmap” fields need little explanation, they represent the main texture and the bump (normal) map used on the model. The “MatCap” field contains a special image that is used to create the lighting of the model. The “Main Color” is used to adjust the blending between unity lighting and matcap lighting.

The best way to understand how these parameters work is by tweaking them. Try some different Matcap images to see how lighting changes (they are located in the “shading textures” folder). The principle is that the normals of the model get their lighting by the pixels of the image, those who face up correspond to the top part of the image, those who face down correspond to the bottom part and so on.

To better understand how “Main Color” adjusts blending between unity lighting and matcap lighting, click on the Main Color swatch. The following window will show up.



The V slider controls the matcap lighting contribution. If you leave it at 125 (the slider is at the middle position), the lighting information from the matcap will influence the model lighting in a one to one scale. If you move the slider to the left, the matcap influence will become weaker and the model darker. Moving the slider to the right, will augment the matcap influence and will make the model brighter.

The A slider controls the unity lights contribution. If you have no lights in your scene, it will be equal to your ambient light settings. In this scene, there is a directional light parented under the main camera. At the middle position it influences the model as normal. If you move the slider to the left or to the right, the model will be less or more influenced from unity lighting correspondingly.

If you leave both sliders at the middle, the model will be equally influenced by unity and matcap lighting. With the A slider at 125 (middle), try to change the directional light x and y rotation (the directional light is parented under the main camera). You will see that cube's lighting changes accordingly.

The H and S sliders can be used to tint the model. Try a blue tint (H = 225, S = 10) for a cold, indoor feel or a yellow tint (H=50, S=10) for a hot, outdoor feel.

Try to add more lights to the scene or to bake a light map. If you have Pro, you can also try out shadows and light probes.

The shaders that are included are the following :

TSD Basic : Matcap only

TSD Detail : Matcap and a Texture

TSD Detail Bump : Matcap, Bump Map and Texture.

TSD Cutout Detail : Matcap and Texture. The Texture alpha controls the opacity of the model.

TSD Cutout Detail Bump : Matcap, Bump Map and Texture. The Texture alpha controls the opacity of the model.

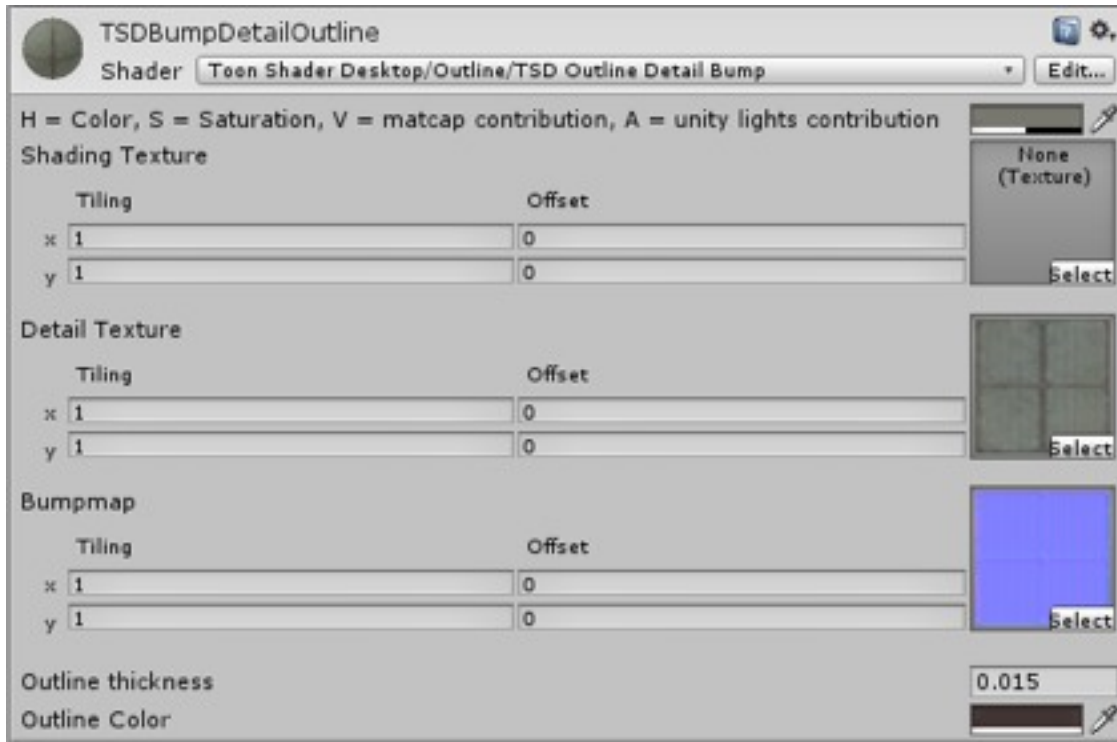
All the above shaders also have an outline version, so they are 10 shaders in total.

TSD Basic and Detail have similar controls with those in the above example.

The Cutout shaders are similar to the unity cutout shaders.

The Outline versions have some more controls. In the folder called “scenes”, open the “Toon Bump Detail Outline” scene. Like before, select the “Toon Cube” in the Hierarchy panel and open the TSDumpDetailOutline in the Inspector panel.

The new elements are located at the bottom.



Outline width controls the thickness of the outlines and outline color sets its color.

Best practices

Models : Use models that are convex, smooth and dense. Uv should be uniform and non overlapping.

Shading textures : Use Texture Type : Advanced, Generate Mip Maps : enabled, Mip Map Filtering : Kaiser, Wrap Mode : Clamp, Filter Mode : Bilinear, Aniso Level : 2 or greater, Format : TrueColor.

Platform : Desktop. For mobile use Toon shader mobile instead.

Contact and acknowledgements

Feel free to contact me about questions, suggestions at unity forums (as Ippokratis), Skype (Bournellis Ippokratis) or at my email : ippobour at gmail dot com.

Many thanks to Texel, Daniel Brauer, Martin Krauss, Jessy Anders, Neil Carter, Michael Garforth, Eric Haines, Keli Hlodversson, Tenebrous, fholm, airship, duckets, Tak, LittleAngel and all the other people who helped me over time to create this shader.

Kind regards,

–Ippokratis